



Adaptive Parameter Selection for Tuning Vision-Language Models

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Abstract

Vision-language models (VLMs) like CLIP have been widely used in various specific tasks. Parameter-efficient fine-tuning (PEFT) methods, such as prompt and adapter tuning, have become key techniques for adapting these models to specific domains. However, existing approaches rely on prior knowledge to manually identify the locations requiring fine-tuning. Adaptively selecting which parameters in VLMs should be tuned remains unexplored. In this paper, we propose CLIP with Adaptive Selective Tuning (CLIP-AST), which can be used to automatically select critical parameters in VLMs for fine-tuning for specific tasks. It opportunely leverages the adaptive learning rate in the optimizer and improves model performance without extra parameter overhead. We conduct extensive experiments on 13 benchmarks, such as ImageNet, Food101, Flowers102, etc, with different settings, including few-shot learning, base-to-novel class generalization, and out-of-distribution. The results show that CLIP-AST consistently outperforms the original CLIP model as well as its variants and achieves state-of-the-art (SOTA) performance in all cases. For example, with the 16-shot learning, CLIP-AST surpasses GraphAdapter and PromptSRC by 3.56% and 2.20% in average accuracy on 11 datasets, respectively. Code will be publicly available.

1. Introduction

With the rise of large-scale image-text pretraining, vision-language models (VLMs) [8, 29, 38, 44, 45, 54] have shown strong capabilities in multimodal representation. These models learn the distribution between visual and textual features by jointly training on large-scale image and text paired data, which enables them to perform well across specific tasks. For example, CLIP [38], trained with hundreds of

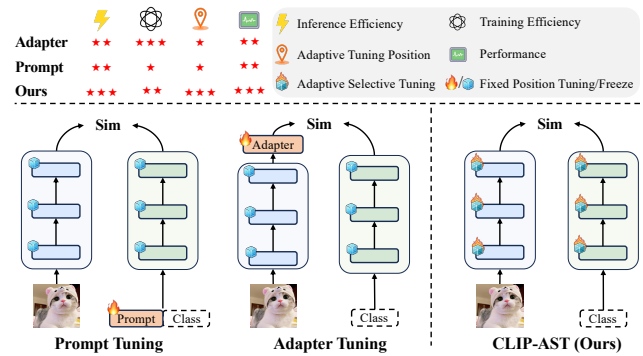


Figure 1. Comparison of fine-tuning methods across key dimensions. Prompt tuning and adapter tuning rely on prior knowledge to define positions for fine-tuning. In contrast, our method performs adaptive selective tuning. It automatically identifies the most critical positions for tuning. Additionally, no extra parameters are introduced, improving efficiency and performance.

millions of image-text pairs using the contrastive learning framework, aligns image and text representations in a high-dimensional feature space, which gives CLIP strong generalization capabilities and makes it widely used in various applications [7, 10, 24, 43, 47, 51, 56, 58, 61].

While CLIP has shown strong general capabilities, its performance often remains suboptimal on specific tasks without task-oriented fine-tuning. However, performing full fine-tuning on large pre-trained models often introduces challenges, including increased computational cost, potential overfitting, and reduced transferability across diverse tasks and domains. Parameter-efficient fine-tuning (PEFT) [2, 4, 15, 27] has been widely adopted to mitigate the computational cost and overfitting risks associated with the full fine-tuning of large pre-trained models.

Although PEFT reduces computational costs and improves results, most existing methods rely on prior knowledge to decide which model components to fine-tune. These methods typically introduce specialized learnable modules at predefined positions in the model. For example, prompt

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tuning [41, 59, 60] incorporates learnable prompt vectors at the input layer or specific transformer layers of CLIP, guiding the model's representation learning. In contrast, adapter tuning [15, 32, 37, 46] inserts additional lightweight modules within the transformer layers or at the final layer of CLIP to adapt to new tasks. However, these methods rely on the manual selection of model locations for fine-tuning, which can be suboptimal and often results in extra parameter overhead. Automatically selecting and fine-tuning the most critical parameters remains underexplored. Accurately selecting critical parameters can effectively address these limitations, thereby enhancing model performance without extra parameter costs.

To tackle this challenge, as illustrated in Fig. 1, we propose CLIP with Adaptive Selective Tuning (CLIP-AST), which aims to automatically select and fine-tune the most important parameters in CLIP without inserting extra learnable modules. CLIP-AST leverages the adaptive learning rate feature of the AdamW optimizer [28], where the effective learning rate of a parameter is inversely proportional to the second-moment estimate of its gradients. Parameters with lower second-moment estimates receive higher effective learning rates, allowing them to adapt quickly to new tasks. Conversely, parameters with higher second-moment estimates receive smaller updates, preventing large adjustments due to noisy gradients. By utilizing this property, we use second-moment estimates as importance scores to select and fine-tune the most critical parameters for the task.

Building on the dynamic learning rate property in the optimizer, CLIP-AST introduces adaptive selective fine-tuning. For each transformer layer, we rank the parameters based on importance scores and fine-tune the top K most important ones while freezing the others. This adaptive selective fine-tuning preserves the knowledge of the pre-trained model. It enhances task-specific adaptation without extra parameters during training and inference.

The main contributions are as follows:

- We propose CLIP-AST, a novel method that automatically selects and fine-tunes the most critical parameters in CLIP without extra parameters, thus maintaining the same inference efficiency as the original CLIP model.
- We introduce an automatic parameter identification strategy based on AdamW's adaptive learning rates to select important parameters with lower second-moment estimates of the gradient.
- Extensive experiments on 13 benchmark datasets demonstrate that CLIP-AST consistently outperforms the original CLIP and its variants, achieving state-of-the-art performance.

2. Related Work

Vision-Language Models. Vision-Language Models (VLMs) [18, 29, 38, 44, 54] aim to efficiently handle

cross-modal tasks by jointly learning visual and linguistic information. Some methods, such as CLIP [38] and ALIGN [18], utilize hundreds of millions or even billions of image-text pairs for training to bridge the gap between visual and textual data. The vast amount of training data enables VLMs to exhibit exceptional capabilities. Among these models, the most representative is the CLIP model, which employs contrastive learning to map images and texts into the same feature space, thereby achieving efficient zero-shot learning. Due to its powerful cross-modal and zero-shot capabilities, CLIP has been widely used in various downstream tasks [10, 25, 30, 36, 43, 48, 51, 56, 58]. However, a gap exists between most downstream tasks and the original CLIP model, necessitating fine-tuning for effective transfer. Therefore, this work investigates the fine-tuning of CLIP, allowing CLIP to be effectively adapted to specific domains while maintaining its strong performance.

Parameter-Efficient Fine-Tuning for VLMs. Recently, with the rapid development of parameter-efficient fine-tuning (PEFT) [15, 16, 27] in the field of NLP, it has become a common paradigm for fine-tuning VLMs. Common PEFT methods include adapters [4, 11, 15, 22, 32, 37, 46], which introduce additional lightweight trainable module, as well as prompt tuning [19, 59, 60], which optimizes input prompts. In the context of applying PEFT to VLMs, CoOp [60] modifies the static text encoder prompts in the original CLIP model into adaptable, trainable vectors. Tip-Adapter [55] develops a key-value caching mechanism to facilitate efficient knowledge retrieval. PromptSRC [21] implements a self-regulation technique to mitigate overfitting. MMA [52] introduces a multimodal adapter to enhance the alignment between text and visual representations. Unlike previous methods, which choose the fine-tuning locations based on prior knowledge, we automatically find the important parameters in the original model that need to be trained and complete the fine-tuning of the model.

Selective Fine-tuning. Unlike approaches that retrain all model weights or add new trainable modules, selective fine-tuning [12, 40, 53, 57] focuses on training specific parameters within the model to achieve fine-tuning. A notable example is linear probing [3, 13, 38], where only the model's last layer is trained. However, restricting training to the last layer often yields suboptimal results. BitFit [53] introduces bias-only fine-tuning, which adjusts only the bias parameters (or a subset of them), keeping the rest of the model parameters fixed. These methods, however, rely on empirical insights or observations to select parameters for training, like the previous PEFT methods for VLMs. In contrast, BSR [40] introduces a systematic selective fine-tuning approach that identifies critical layers by comparing outputs from pre-trained and fine-tuned models. Layers showing significant output changes are prioritized for

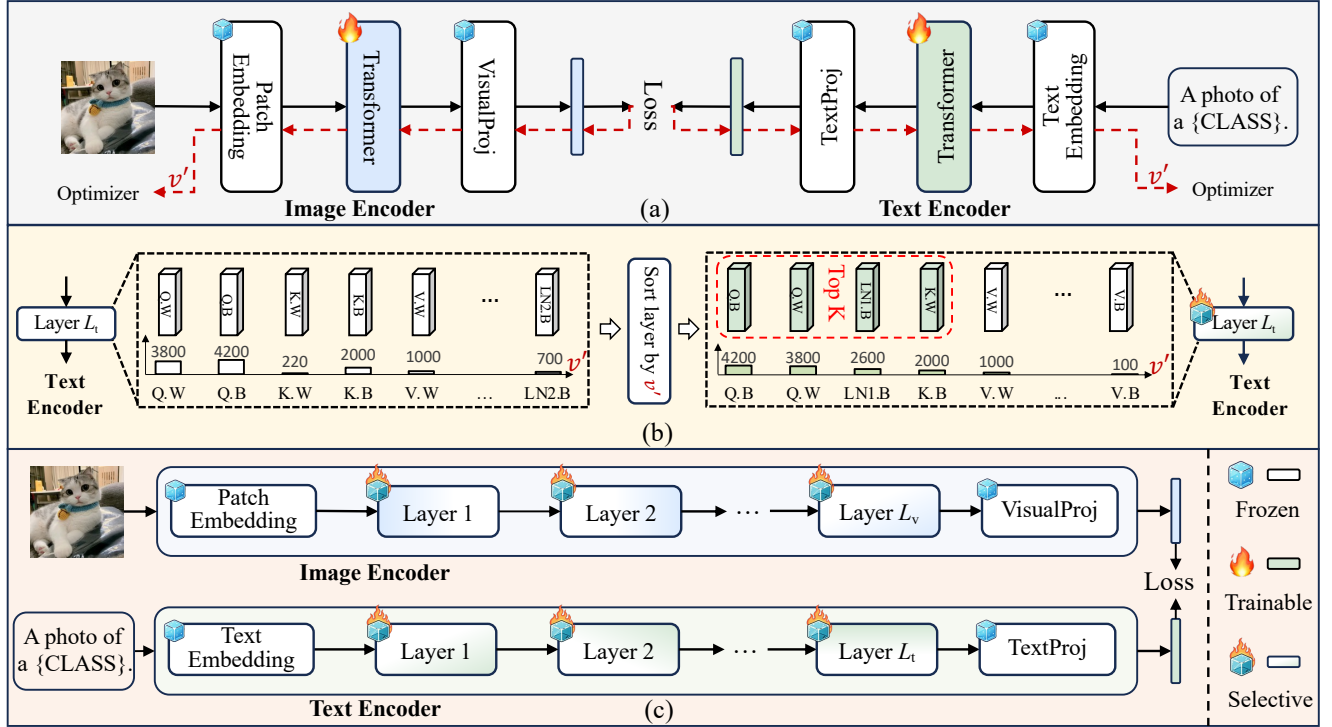


Figure 2. Overview of the proposed CLIP-AST pipeline, consisting of (a) transformer fine-tuning and (b) (c) adaptive selective fine-tuning stages. In (a), we fine-tune the transformer layers of both visual and text encoders to extract parameter importance scores. (b) illustrates the process of ranking sub-layer parameters using importance scores, followed by the adaptive selection of the top K sub-layers. (c) shows the selective fine-tuning process where only the most important sub-layers are updated while other components remain frozen, enhancing efficiency without compromising model performance.

fine-tuning based on task adaptability. SPT [12] takes a data-driven approach by computing parameter sensitivity on downstream tasks. It identifies the most impactful parameters and employs structured fine-tuning methods (e.g., LoRA or Adapter) rather than direct adjustments. Despite these advances, adaptive selective fine-tuning for CLIP remains relatively unexplored. This work addresses this gap by introducing an automated method for identifying the modules in CLIP that benefit most from fine-tuning, advancing the efficiency and effectiveness of fine-tuning for vision-language models.

3. Method

As illustrated in Fig. 2, CLIP-AST includes two stages: transformer fine-tuning and adaptive selective fine-tuning. Initially, we fine-tune the pre-trained layers of the transformer to extract importance scores from the optimizer. The optimizer measures the importance of parameters by estimating the second-order moments of the gradients in each transformer layer, as described in Sec. 3.1. After a few iterations, these scores are ranked to select the parameters requiring training, as specified in Sec. 3.2. Then, the important parameters in each layer of the transformer are fine-

tuned to complete the fine-tuning of CLIP.

3.1. Transformer Fine-Tuning

To assess the importance of all the parameters of the transformer layer in the pre-trained CLIP model, as shown in Fig. 2 (a), we first fine-tune the CLIP transformer block. The CLIP model includes a visual encoder E_v and a textual encoder E_t , both responsible for mapping images and text into a unified feature space. This design facilitates the interaction of features across different modalities. During the fine-tuning of CLIP's transformer, we first sampled a batch of training data from the dataset, which included N images $\mathbf{X} \in \mathbb{R}^{N \times 3 \times H \times W}$, corresponding labels $\mathbf{Y} \in \mathbb{R}^N$, and the names of C classes $\mathbf{T} = \{\mathbf{T}_k\}, k \in \{1, \dots, C\}$ in the dataset. Subsequently, the visual encoder E_v extracts visual embeddings $\mathbf{G} \in \mathbb{R}^{N \times D}$ from the $\mathbf{X}$:

$$\begin{aligned} [\mathbf{G}_0, \mathbf{I}_0] &= \text{PatchEmbedding}(\mathbf{X}), \\ [\mathbf{G}_L, \mathbf{I}_L] &= \text{Transformer}([\mathbf{G}_0, \mathbf{I}_0]), \\ [\mathbf{G}, \mathbf{I}] &= \text{VisualProj}([\mathbf{G}_L, \mathbf{I}_L]), \end{aligned} \quad (1)$$

where $\mathbf{G}_0$ and $\mathbf{I}_0$ represent the initial global and visual embeddings generated by the PatchEmbedding block from the input visual $\mathbf{X}$. The Transformer block processes these

embeddings through multiple layers to refine the features, resulting in $\mathbf{G}_L$ and $\mathbf{I}_L$, which are the final global and visual embeddings at the last layer. Finally, the VisualProj block projects these embeddings onto a feature space, producing $\mathbf{G}$ and $\mathbf{I}$ as the final visual embeddings.

For the text encoder E_t , the class names T are combined with a predefined template (e.g., "a photo of [CLASS]") to generate class descriptions $\mathbf{T}_p$. Then, each class description is sent to the text encoder to obtain the corresponding class embeddings $\mathbf{W} \in \mathbb{R}^{C \times D}$:

$$\begin{aligned}\mathbf{W}_0 &= \text{TextEmbedding}(\mathbf{T}_p), \\ \mathbf{W}_L &= \text{Transformer}(\mathbf{W}_0), \\ \mathbf{W} &= \text{TextProj}(\mathbf{W}_L).\end{aligned}\quad (2)$$

Here, $\mathbf{W}_0$ represents the initial text embeddings generated by the TextEmbedding layer. In contrast, $\mathbf{W}_L$ is the feature produced by passing $\mathbf{W}_0$ through multiple layers of the transformer, extracting semantic information further at each layer. Finally, TextProj projects the text features into a unified feature space, producing $\mathbf{W}$ as the final text embedding that aligns with the visual features.

After obtaining the $\mathbf{G}$ and $\mathbf{W}$, we first calculate the probability $\hat{y}_{i,j}$ of each image being predicted for each class and then calculate the loss function with the label:

$$\begin{aligned}\hat{y}_{i,j} &= \frac{\exp(\text{sim}(\mathbf{G}_i, \mathbf{W}_j)/\tau)}{\sum_{k=1}^C \exp(\text{sim}(\mathbf{G}_i, \mathbf{W}_k)/\tau)}, \\ \mathcal{L}_{CE} &= -\frac{1}{N} \sum_{j=1}^N \sum_{i=1}^C y_{i,j} \log(\hat{y}_{i,j}),\end{aligned}\quad (3)$$

where $\text{sim}(\cdot, \cdot)$ represents the cosine similarity function, τ is a temperature parameter.

Fine-tuning the transformer requires only a few training iterations. As its name implies, only the transformer part of the two encoders is trainable during the training process. After training, we can obtain parameter importance scores v , which are embedded within the AdamW optimizer.

3.2. Adaptive Selective Fine-tuning

After transformer fine-tuning, we introduce the adaptive selective fine-tuning strategy, which uses the square root of the second-moment estimate of the gradient in the AdamW optimizer [28] as importance scores. The parameter update process in the i -th iteration of AdamW optimization can be described as follows. First, we compute the gradient g_i with respect to the model parameters θ :

$$g_i = \nabla_{\theta} f(\theta_{i-1}). \quad (4)$$

Next, we update the first-moment estimate m_i and the second-moment estimate v_i :

$$\begin{aligned}m_i &= \beta_1 \cdot m_{i-1} + (1 - \beta_1) \cdot g_i, \\ v_i &= \beta_2 \cdot v_{i-1} + (1 - \beta_2) \cdot g_i^2,\end{aligned}\quad (5)$$

where β_1 and β_2 are the hyperparameters controlling the exponential decay rates for the moment estimates. To correct the bias introduced in the early stages of optimization, we compute the bias-corrected estimates:

$$\hat{m}_i = \frac{m_i}{1 - \beta_1^i}, \quad \hat{v}_i = \frac{v_i}{1 - \beta_2^i}. \quad (6)$$

The final parameter update rule is then expressed as:

$$\theta_i = (\theta_{i-1} - \alpha \cdot \lambda \cdot \theta_{i-1}) - \frac{\alpha}{\sqrt{\hat{v}_i} + \epsilon} \cdot \hat{m}_i, \quad (7)$$

where α denotes the learning rate, λ represents the weight decay coefficient that controls the strength of regularization applied to the model parameters, and ϵ is a small constant added for numerical stability to prevent division by zero.

In our adaptive selective fine-tuning, the importance of each parameter is assessed using the square root of the second-moment estimate of gradient $\hat{v}_i$, which has the same dimensionality as the parameter itself. Specifically, we compute the scalar average importance score v' of the transformer layer sub-layer parameter i as follows:

$$v'_i = \text{Avg} \left(\frac{1}{\sqrt{\hat{v}_i}} \right). \quad (8)$$

This score measures the magnitude of parameter updates during training. A higher v'_i means the parameter receives larger updates, suggesting it is more actively involved in learning and adapting to the task-specific data. Such parameters will likely capture essential features and contribute significantly to model predictions. Conversely, a lower v'_i implies that the parameter receives smaller updates, possibly because it requires less adjustment or is less sensitive to the current data.

By focusing on parameters with higher importance scores, we can adaptively and selectively fine-tune the model, prioritizing the most influential for the specific task. This selective fine-tuning strategy maintains model performance while reducing computational costs, as it concentrates updates on the most critical components without compromising the overall model integrity.

During the adaptive selective fine-tuning, shown in Fig. 2 (b) and (c), we select the most important components at the sub-layer level (e.g., weights or biases of the linear_q layer). For each layer in the transformer module, we select the top K important parameters of the sub-layer for training while freezing all others. This adaptive selective fine-tuning strategy enhances the efficiency and effectiveness of the model, allowing for focused updates that improve task-specific adaptation without compromising overall model integrity.

3.3. Loss Function

We utilize the self-consistency loss (SCL) [21] to mitigate overfitting during selective fine-tuning. SCL employs an additional frozen CLIP model to extract original visual embeddings $\tilde{\mathbf{G}}$ and textual embeddings $\tilde{\mathbf{W}}$ from the training data. By using an L1 loss, SCL enforces consistency between the features produced by the frozen model and the features produced by the fine-tuned model:

$$\begin{aligned}\mathcal{L}_{\text{SCL-image}} &= \sum_{i=1}^d |\tilde{\mathbf{G}}_i - \mathbf{G}_i|, \\ \mathcal{L}_{\text{SCL-text}} &= \sum_{i=1}^d |\tilde{\mathbf{W}}_i - \mathbf{W}_i|.\end{aligned}\quad (9)$$

In addition, SCL introduces logit-level consistency regularization, further strengthening the constraints and maximizing the alignment of features before and after training. This is achieved by minimizing the Kullback-Leibler divergence to align the trained logit distribution with the logits generated by the frozen CLIP model, as expressed in the following formula:

$$\mathcal{L}_{\text{SCL-logits}} = \mathcal{D}_{\text{KL}}(\text{sim}(\tilde{\mathbf{G}}, \tilde{\mathbf{W}}), \text{sim}(\mathbf{G}, \mathbf{W})). \quad (10)$$

Finally, we incorporate the cross-entropy loss discussed in Sec. 3.1. The total loss our model aims to minimize is a combination of these types of losses, represented by the equation:

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \lambda_1 \cdot \mathcal{L}_{\text{SCL-image}} + \lambda_2 \cdot \mathcal{L}_{\text{SCL-text}} + \lambda_3 \cdot \mathcal{L}_{\text{SCL-logits}}, \quad (11)$$

where λ_1 , λ_2 and λ_3 are weights that balance the contributions of $\mathcal{L}_{\text{SCL-image}}$, $\mathcal{L}_{\text{SCL-text}}$ and $\mathcal{L}_{\text{SCL-logits}}$, respectively.

4. Experiments

4.1. Experimental Settings

Few-shot learning. We aim to evaluate the model's ability to generalize with limited labeled examples in the few-shot learning setting. We simulate a low-shot scenario by providing only a few labeled samples per class, typically 1, 2, 4, 8, 16. In this setting, we use the 11 commonly used classification datasets introduced in Sec. 4.2 to reflect the few-shot tasks in the real world. The performance under this setting is measured by top-1 accuracy.

Base-to-novel class generalization. The base-to-novel class generalization setting evaluates the model's adaptability to novel classes it has not seen during fine-tuning. In this setting, the model is trained on a base set of classes with 16-shot per class and is then tested on novel classes, assessing its capability to transfer knowledge from the base classes to novel, unseen classes. Following the previous work [21, 60], we partition the dataset into base and novel

classes, ensuring no overlap between the two. The metrics include the top-1 accuracy of the base class (Base) and the novel class (Novel), as well as the harmonic mean (HM) of the two.

Out-of-distribution setting. In the out-of-distribution setting, we test the model's robustness against data that differs from the training distribution. This setting aims to assess the model's ability to handle unexpected inputs. We train on ImageNet as an in-distribution dataset and test on out-of-distribution datasets. The performance is evaluated for average accuracy on both in-distribution and out-of-distribution samples.

4.2. Datasets

For few-shot learning and base-to-novel class generalization, we conduct experiments on 11 commonly used classification benchmarks, including Caltech101 [9], DTD [5], EuroSAT [14], FGVC Aircraft [31], Flowers102 [33], Food101 [1], ImageNet [6], OxfordPets [34], StanfordCars [23], SUN397 [50], UCF101 [42]. For the out-of-distribution setting, We use ImageNet-Sketch [49] and ImageNetV2 [39] as out-of-distribution datasets.

4.3. Implementation details

For the adaptive selective fine-tuning stage, the preprocessing steps during training include randomly cropping the input images, with the crop size ranging from 0.5 to 1.0 times the original image size. The cropped images are then resized to a fixed size of 224×224 pixels. Next, each image undergoes horizontal flipping with a probability of 0.5. Finally, each channel of the images is normalized. We use the AdamW optimizer for training. For the few-shot learning and out-of-distribution settings, we train for 30 epochs, and for the base-to-novel class generalization setting, we train for 20 epochs. To ensure a fair comparison, we follow previous methods [21, 52] and use the ViT-B/16 CLIP model. For the transformer fine-tuning stage, we train for 1 epoch, with other settings being the same as adaptive selective fine-tuning. All experiments are implemented using the Jittor [17] and PyTorch [35] frameworks.

4.4. Comparison with State-of-the-art

Few-shot learning. In this setting, we benchmark CLIP-AST against state-of-the-art (SOTA) few-shot learning methods, including CoOp [60], Tip-Adapter [55], PromptSRC [21], and GraphAdapter [26]. As shown in Tab. 1, CLIP-AST consistently achieves the highest average accuracy across the ImageNet dataset as the shot count increases, demonstrating that our proposed method is on par with, or even surpasses, previously introduced methods such as prompt tuning and adapters. To further clarify the comparison, we present the overall and detailed results for 11 datasets at varying shot levels in Fig. 3. Notably, CLIP-

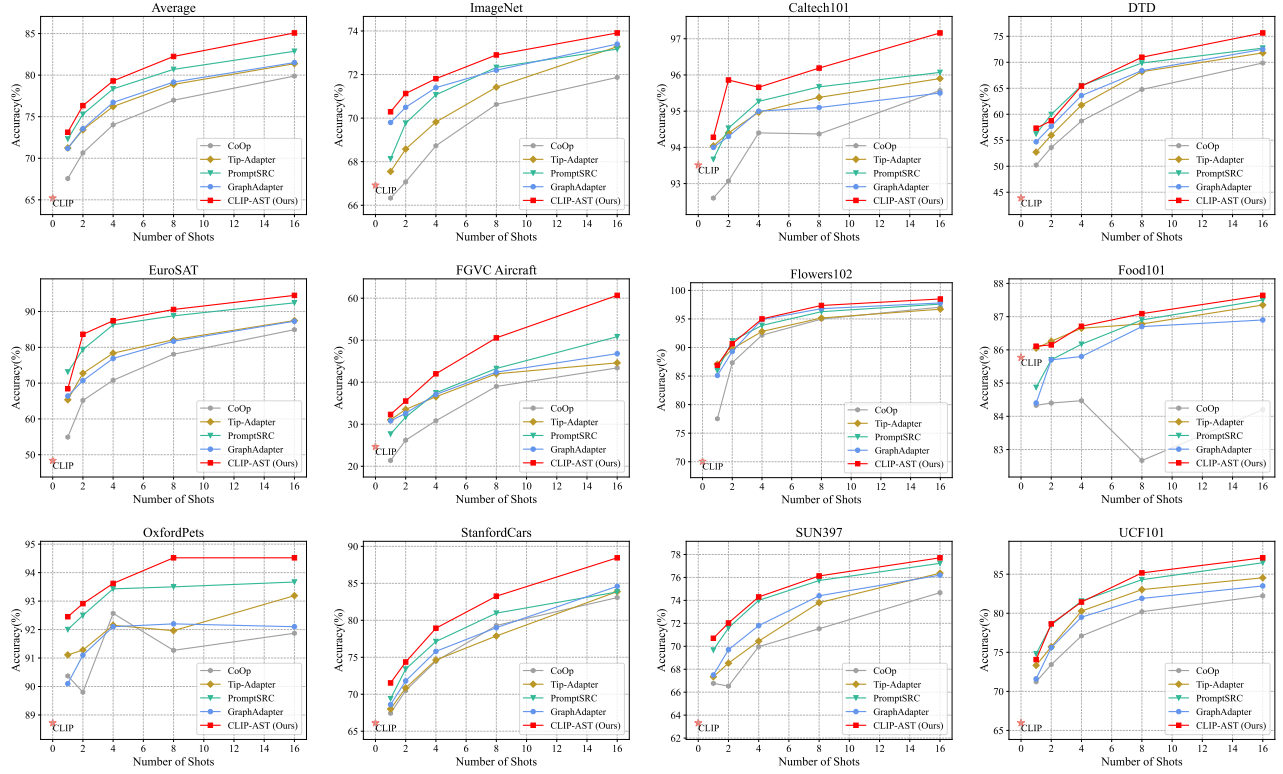


Figure 3. Comparison with previous SOTA across 11 datasets under 1, 2, 4, 8, and 16-shot settings demonstrates that our approach consistently outperforms existing methods, setting a new SOTA.

AST achieves an average accuracy improvement of 2.21% over the SOTA across 11 datasets. Across all datasets, our method outperforms previous SOTA approaches, with particularly substantial improvements on certain datasets. For instance, on FGVC Aircraft, CLIP-AST achieves approximately 10% higher accuracy with 16-shot, surpassing other methods such as PromptSRC and CoOp, which are specifically designed for prompt optimization. Moreover, as the sample count increases, the performance advantage of our model over prior methods also becomes more obvious. The consistent progress achieved by CLIP-AST on different levels under different samples demonstrates that our proposed fine-tuning method is a simple yet highly effective strategy in the few-shot learning setting.

Base-to-novel class generalization. In this setting, we evaluate the ability of CLIP-AST to generalize from base to novel classes by comparing it with state-of-the-art methods in Tab. 2, where CLIP-AST consistently performs favorably in both base and novel class recognition tasks. CLIP-AST achieves the highest harmonic mean (HM) across multiple datasets, indicating robust performance in both base and novel classes. For example, on the EuroSAT dataset, our method achieves an HM improvement of approximately 4.37% compared to MMA. This demonstrates the capabil-

Method	Number of Shot				
	1	2	4	8	16
CLIP [38]	Zero-shot 66.92				
CoOp [60]	66.33	67.07	68.73	70.63	71.87
CoCoOp [59]	69.43	69.78	70.39	70.63	70.83
Tip-Adapter [55]	67.55	68.58	69.82	71.42	73.28
PromptSRC [21]	68.13	69.77	71.07	72.33	73.17
GraphAdapter [26]	69.80	70.50	71.40	71.40	73.40
CLIP-AST (Ours)	70.29	71.13	71.81	72.90	73.91

Table 1. Compared with the previous SOTA methods in the ImageNet dataset with the few-shot learning setting.

ity of CLIP-AST to maintain high accuracy across varying class distributions without overfitting to base classes. Our results illustrate that selective fine-tuning in a base-to-novel setting allows CLIP-AST to balance learning across class distributions effectively. By avoiding overfitting to base classes, our approach maintains strong generalization to novel classes. The gains across both base and novel categories prove that CLIP-AST can manage the trade-off between base and novel class performance.

Out-of-distribution setting. In this setting, we assess the out-of-distribution robustness of CLIP-AST by comparing it with leading methods in Tab. 3. CLIP-AST achieves su-

Dataset		CLIP [38]	CoCoOp [59]	MaPLe [20]	PromptSRC [21]	MMA [52]	CLIP-AST (Ours)
Average	Base	69.34	80.47	82.28	84.26	83.20	85.94
	Novel	74.22	71.69	75.14	76.10	76.80	76.99
	HM	71.70	75.83	78.55	79.97	79.87	81.06
Caltech101	Base	96.84	97.96	97.74	98.10	98.40	98.71
	Novel	94.00	93.81	94.36	94.03	94.00	94.00
	HM	95.40	95.84	96.02	96.02	96.15	96.30
DTD	Base	53.24	77.01	80.36	83.37	83.20	84.03
	Novel	59.90	56.00	59.18	62.97	65.63	65.34
	HM	56.37	64.85	68.16	71.75	73.38	73.52
EuroSAT	Base	56.48	87.49	94.07	92.90	85.46	95.90
	Novel	64.05	60.04	73.23	73.90	82.34	81.72
	HM	60.03	71.21	82.35	82.32	83.87	88.24
FGVC Aircraft	Base	27.19	33.41	37.44	42.73	40.57	48.98
	Novel	36.29	23.71	35.61	37.87	36.33	38.21
	HM	31.09	27.74	36.50	40.15	38.33	42.93
Flowers102	Base	72.08	94.87	95.92	98.07	97.77	97.91
	Novel	77.80	71.75	72.46	76.50	75.93	77.73
	HM	74.83	81.71	82.56	85.95	85.48	86.66
Food101	Base	90.10	90.70	90.71	90.67	90.13	90.57
	Novel	91.22	91.29	92.05	91.53	91.30	91.11
	HM	90.66	90.99	91.38	91.10	90.71	90.84
ImageNet	Base	72.43	75.98	76.66	77.60	77.31	78.44
	Novel	68.14	70.43	70.54	70.73	71.00	70.22
	HM	70.22	73.10	73.47	74.01	74.02	74.10
OxfordPets	Base	91.17	95.20	95.43	95.33	95.40	96.23
	Novel	97.26	97.69	97.76	97.30	98.07	97.37
	HM	94.12	96.43	96.58	96.30	96.72	96.80
Stanford Cars	Base	63.37	70.49	72.94	78.27	78.50	84.21
	Novel	74.89	73.59	74.00	74.97	73.10	74.05
	HM	68.65	72.01	73.47	76.58	75.70	78.80
SUN397	Base	69.36	79.74	80.82	82.67	82.27	83.05
	Novel	75.35	76.86	78.70	78.47	78.57	78.12
	HM	72.23	78.27	79.75	80.52	80.38	80.51
UCF101	Base	70.53	82.33	83.00	87.10	86.23	87.38
	Novel	77.50	73.45	78.66	78.80	80.03	79.12
	HM	73.85	77.64	80.77	82.74	82.20	83.05

Table 2. Compared with the previous SOTA methods in the base-to-novel class generalization setting, where the model is trained on the base class with 16-shot and evaluated on the base class and novel class.

perior performance on both in-distribution (ImageNet) and out-of-distribution target datasets (ImageNet-Sketch and ImageNet-V2). CLIP-AST achieves the highest average accuracy across these datasets, notably achieving a 2.04% improvement over the next-best method on ImageNet-V2. This robust performance on challenging out-of-distribution tasks highlights the strength of our method for maintaining model generalization beyond the source distribution. These results demonstrate that CLIP-AST can effectively mitigate the domain shift, balancing in-distribution and out-of-distribution accuracy.

4.5. Ablation Study

In our ablation study using the few-shot learning setting, we first analyzed our selected approach’s training and inference efficiency compared to previous fine-tuning paradigms. Then, we conducted an ablation study across 11 datasets. We extensively evaluated different approaches to selective fine-tuning and then performed a detailed ablation test on the hyperparameters of the CLIP-AST.

Training and inference efficiency analysis. We compare the training and inference efficiency of various methods in Tab. 4. Our method achieves the highest accuracy of 85.64% while maintaining competitive training and infer-

Method	Source ImageNet	Target		Avg Acc(%)
		-Sketch	-V2	
CLIP [38]	66.73	46.15	60.83	57.90
CoOp [60]	71.51	47.99	64.20	61.23
CoCoOp [59]	71.02	48.75	64.07	61.28
Tip-Adapter [55]	73.23	46.82	65.01	61.68
MaPLe [20]	70.72	49.15	64.07	61.31
PromptSRC [21]	71.27	49.55	64.35	61.72
MMA [52]	71.00	49.13	64.33	61.48
CLIP-AST (Ours)	73.87	48.37	66.37	62.87

Table 3. Compared with the previous SOTA methods in the out-of-distribution setting, where the model is trained on the ImageNet dataset with 16-shot and evaluated on the ImageNet-V2 and ImageNet-Sketch benchmarks.

Method	Tuning Type	Train time(s)	Inference FPS	Acc(%)
CLIP [38]	-	0	1323	66.12
CoCoOp [59]	Prompt	2280	54	72.00
Tip-Adapter [55]	Adapter	30	1280	83.09
PromptSRC [21]	Prompt	630	1236	74.70
MMA [52]	Adapter	135	1113	83.60
CLIP-AST (Ours)	Adaptive Selection	100	1323	85.64

Table 4. Comparison of training time, inference efficiency, and accuracy across different methods. All methods are compared using a single NVIDIA 4090 GPU, trained for 10 epochs on the StanfordCars dataset.

ence efficiency. All methods were trained for 10 epochs on the StanfordCars dataset. Our method requires only 100 seconds of training time, less than needed for prompt tuning methods. The inference speed is also the same as the original CLIP model and faster than both prompt and adapter tuning methods. This indicates that our selective fine-tuning method improves performance and maintains high efficiency during the training and inference stages.

Adaptive selective fine-tuning schemes. Our ablation study, summarized in Tab. 5, evaluates the impact of different selective fine-tuning schemes on model performance. We start with a baseline where neither the image encoder nor the text encoder is fine-tuned, resulting in an average accuracy of 65.21%. Fine-tuning only the image encoder or only the text encoder using the adaptive selection strategy yields average accuracies of 84.04% and 81.95%, respectively, indicating significant contributions from both encoders. Next, We compared BitFit [53], which utilizes a priori fixed selection of bias as a training parameter, and found that the method performs poorly. We also compared the global granularity for adaptive selection, where the top K important types of sub-layers are trained across the entire transformer, with the layer granularity selection, where the top K important layers are trained within each layer. These results highlight the effectiveness of finer-grained layer granularity selection.

Hyperparameters in adaptive selection. In Fig. 4, we per-

Trainable Image Encoder	Trainable Text Encoder	Scheme for Selecting Trainable Layers	Avg Acc(%)
✗	✗	-	65.21
✓	✗	layer granularity	84.04
✗	✓	layer granularity	81.95
✓	✓	fixed bias	79.96
✓	✓	global granularity	84.25
✓	✓	layer granularity	85.05

Table 5. Ablation of selective fine-tuning schemes. Fixed bias trains the bias parameters, global granularity chooses the most important top K sub-layers across the model, and layer granularity picks the most important top K sub-layers within each transformer layer. Avg Acc is the average accuracy across 11 datasets in the few-shot learning setting with 16-shot.

formed an ablation on the hyperparameters of adaptive selection. First, we ablated the number of epochs for transformer fine-tuning used to obtain the importance scores and found that the number of epochs had little impact on the results. Next, we ablated the number of training layers K . We found that smaller or larger values of K did not lead to optimal results, possibly due to underfitting and overfitting. Ultimately, we chose 6 sub-layers as the hyperparameter for the few-shot learning setting.

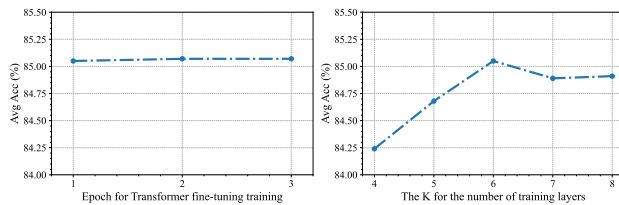


Figure 4. Ablation of epoch for transformer fine-tuning and the number of training layers in adaptive selective fine-tuning.

4.6. Visualization

Statistics of training parameter selection. In Fig. 5, we visualize the parameter selection results in the few-shot setting for 16-shot across 11 datasets. Firstly, we observe that the sub-layers chosen for both the image and text encoders are similar but not identical. Notably, some sub-layers are consistently selected across different layers, indicating that these sub-layers might be responsible for extracting general features. Additionally, the sub-layers selected by the transformer layer are also different, indicating that the functions responsible for shallow and deep layers are not the same.

Image feature analysis before and after training. In Fig. 6, we visualize the distribution of image features extracted by the model before and after training using T-SNE [38]. On the left, the features are scattered with significant overlap between different classes, indicating that the model has not yet effectively distinguished visual features. After training (right side), the features form compact and distinct clusters, demonstrating that the model has successfully learned to extract discriminative features.

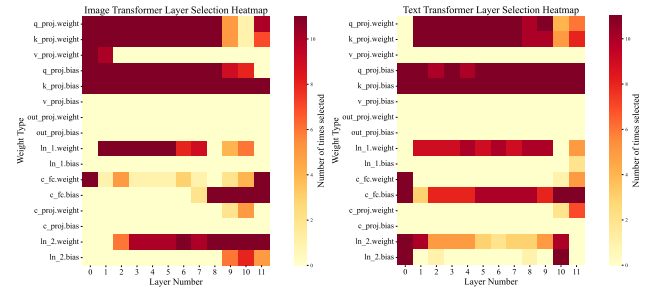


Figure 5. Heatmaps of adaptive parameter selection for image (left) and text (right) encoders across 11 datasets in the 16-shot setting.

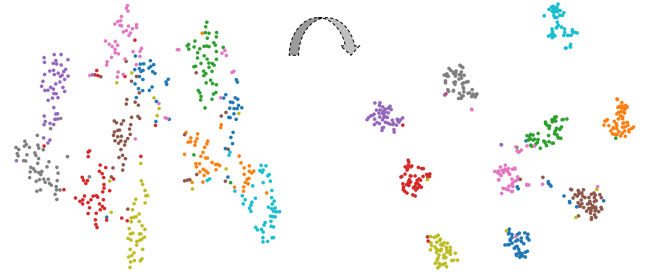


Figure 6. T-SNE visualization of image features before (left) and after (right) training. After training, features form clearer clusters, showing improved class separation.

5. Conclusion

We introduce CLIP-AST, an adaptive selective fine-tuning method for VLMs that addresses the limitations of existing PEFT approaches for CLIP. Unlike traditional methods that rely on the manual selection of adaptation positions, CLIP-AST uses the adaptive learning rate mechanism of the AdamW optimizer to automatically select critical parameters with lower second-moment estimates of their gradients. This enables selective fine-tuning of key sub-layers, preserving pre-trained knowledge, reducing overfitting, and minimizing computational costs. Extensive experiments show that CLIP-AST outperforms the original CLIP model and achieves exceptional performance across numerous benchmarks. Overall, it provides a robust framework for efficient fine-tuning of VLMs, with potential for application in other multimodal architectures and further enhancements in parameter selection strategies. Future work could explore extending this approach to other multimodal architectures and further refine the parameter selection criteria for even greater performance gains.

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